

Development and Evaluation of an Arduino-Based Breathalyzer with MQ-3 Sensor for Breath Alcohol Concentration Estimation

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Abstract— This project report details the development of a low-cost breathalyzer using an Arduino Uno microcontroller and an MQ-3 alcohol sensor. The primary objective was to design, implement, and evaluate a functional system capable of estimating breath alcohol concentration (BrAC) and, subsequently, blood alcohol concentration (BAC). The methodology involved setting up the hardware components, programming the Arduino to read and process the sensor data, and conducting preliminary experiments with two participants. Key findings indicate the feasibility of using the MQ-3 sensor for relative BrAC estimation, though accuracy for precise BAC determination requires careful calibration and consideration of individual physiological factors. The report further discusses the signal processing techniques employed as well as the mathematical explanations behind the calculations of the estimated outputs. Additionally, the report will also justify the design choices in terms of the hardware circuit, microcontroller, and LabView designs, and analyze the findings. Finally, the report concludes by highlighting the potential of this project as a foundational step towards accessible alcohol sensing technology while acknowledging its limitations for critical applications.

I. INTRODUCTION

Breathalyzers are devices used to estimate blood alcohol concentration (BAC) from a breath sample. They play a crucial role in various applications, ranging from law enforcement to healthcare and personal safety. By providing a non-invasive method for alcohol detection, breathalyzers contribute significantly to reducing alcohol-related accidents and promoting responsible alcohol consumption. While professional-grade breathalyzers often employ sophisticated technologies, there is a growing interest in exploring accessible and low-cost alternatives for personal use and educational purposes. This project aims to address this interest by designing, implementing, and evaluating a functional breathalyzer based on the readily available Arduino Uno microcontroller and the MQ-3 alcohol sensor. The core principle behind this project lies in the MQ-3 sensor's ability to detect alcohol vapor and

the subsequent estimation of BAC using established methods.

The MQ-3 alcohol sensor is a semiconductor gas sensor specifically designed for detecting alcohol (ethanol) vapor. Its sensing material is tin dioxide (SnO_2), a semiconductor that exhibits lower electrical conductivity in clean air. When alcohol vapor is present in the environment, it reacts with the surface of the tin dioxide, causing a change in its electrical conductivity. Specifically, the conductivity of the sensor increases (which translates to a decrease in resistance) as the concentration of alcohol vapor rises. This change in resistance is proportional to the alcohol concentration in the air and forms the basis for the sensor's detection capabilities [7]. The sensor incorporates an internal heater element, which is crucial for its operation and proper response to alcohol vapor. This heater typically requires a 5V DC power supply to reach the necessary operating temperature. The fundamental working principle relies on the inverse relationship between alcohol concentration and the sensor's resistance. This relationship needs to be accurately captured and processed to estimate the breath alcohol concentration [9].

To estimate blood alcohol concentration (BAC), the Widmark formula is often employed. This formula is a widely used method for estimating BAC based on the amount of alcohol consumed, an individual's body weight, and their sex, which influences the volume of distribution of alcohol in the body. A simplified version of the formula can be expressed as [4]:

$$BAC = \frac{\text{Alcohol (grams)}}{\text{Body Weight (kg)} \times \text{Distribution Factor}} \times 100\%$$

The distribution factor accounts for the proportion of body weight that is water, which

varies between individuals and typically differs between men and women. While the Widmark formula provides a useful estimation, it is important to recognize its limitations. It is a statistical model that relies on averages and does not account for all the individual physiological differences that can affect how alcohol is metabolized and distributed in the body. Factors such as metabolism rate, food intake, and individual variations in body composition can lead to deviations between the estimated BAC and the actual BAC level. Therefore, while the Widmark formula is a standard tool for estimation, its inherent limitations, particularly concerning individual variability, must be considered when interpreting the results from a DIY breathalyzer [4].

To address the limitations of the Widmark formula and improve the reliability of BAC estimation, this project also incorporates sensor-based measurements of breath alcohol concentration (BrAC) using the MQ-3 gas sensor. BrAC provides a real-time approximation of alcohol vapor concentration in exhaled breath, and although it does not directly measure BAC, it can be used to estimate it through a standardized conversion. A commonly accepted ratio in forensic toxicology is that 2100 mL of alveolar air contains the same amount of ethanol as 1 mL of blood, allowing BrAC values (in mg/L) to be multiplied by 2.1 to yield an estimated BAC (%) [3]. This creates a second, independent estimate of BAC that reflects real-time physiological data. By comparing the BAC derived from the Widmark formula with the sensor-derived BAC (BrAC X 2.1), we establish a range of plausible values that accounts for both user-specific input and real-time breath alcohol levels. This dual-estimation method highlights the value of sensor integration in DIY breathalyzers, providing a more nuanced and more accurate representation of intoxication.

II. MATERIALS AND METHODS

As depicted in Figure 1 the hardware components utilized in this project include an Arduino Uno microcontroller, which was connected to a PC via a USB to Type-B, an MQ-3 alcohol sensor module, connecting wires, and resistors; specifically, 6

resistors connected in series to create 200k. These components were connected on a breadboard to create the prototype circuit. The MQ-3 alcohol sensor used in this project features a 6-pin configuration, which includes two heater pins (H) and four signal-related pins (A and B). In this setup, pins A and B are electrically equivalent pairs used to measure the resistance across the sensor (R_s). As shown in the schematic, a voltage divider circuit is formed by connecting one A-B pair to 5V (VCC), and the other to a load resistor (R_L) to ground. The junction between R_S and R_L produces an analog voltage output (V_O) that is read by the Arduino's analog pin (A5). This voltage varies with the alcohol vapor concentration in the air, allowing estimation of the sensor resistance and thus the breath alcohol concentration [9]. The Arduino supplies 5V to both the heater and the sensing circuit, simplifying integration. This configuration ensures accurate and granular analog readings for real-time estimation of breath alcohol levels.

In the Arduino code, the analog voltage from the sensor was read using the `analogRead()` function, which returns a value between 0 and 1023 corresponding to an input voltage between 0 and 5 V. This value was converted to an actual voltage using the equation:

$$V_{out} = \left(\frac{\text{analog reading}}{1023} \right) \times 5.0$$

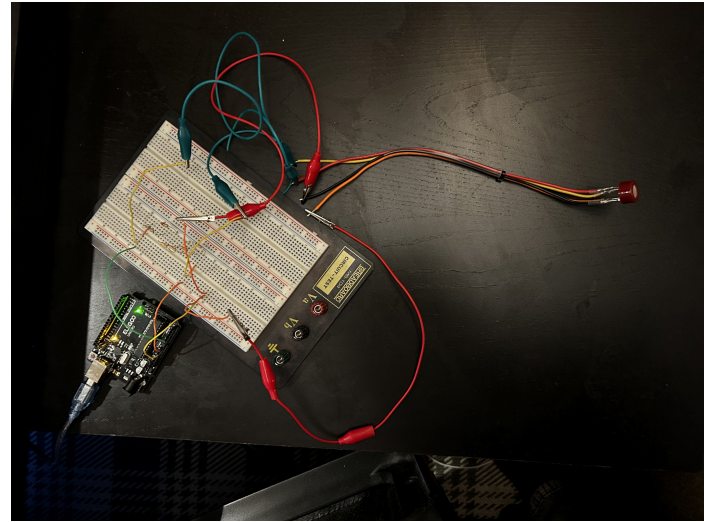


Figure 1: Circuit configuration

The resistance of the sensor in the presence of alcohol, denoted as R_S , was then calculated using Ohm's law, based on the voltage divider

configuration of the circuit. The load resistor R_L used in this setup was 200 k Ω . The sensor resistance was calculated as:

$$R_s = \left(\frac{V_{cc} \times R_L}{V_{out}} \right) - R_L$$

Where VCC is the supply voltage (5 V) and VOut is the measured sensor output voltage.

To estimate the breath alcohol concentration (BrAC) in mg/L, the sensor must first be calibrated by determining its resistance in clean air (R_0). The sensor must then be calibrated in order to find the relationship between BrAC and RSRO. The MQ-3 sensor is known to follow a power-law relationship, as evidenced by the log-log scale used in its datasheet sensitivity curve. This is apparent from the exponentially spaced grid lines on both axes, indicating that both the gas concentration and the resistance ratio RSRO are plotted on logarithmic scales. On such a scale, a power-law relationship appears as a straight line — a pattern observed in the datasheet, where the alcohol curve specifically follows a straight-line trajectory — and corresponds to the equation:

$$\log\left(\frac{R_s}{R_0}\right) = \log(a) + (b \times \log(Brac))$$

From this relationship, the slope of the line represents b , while the intercept provides the value of $\log(a)$. Finding the values of these variables will help us determine a relationship between the BrAC and the resistance ratio. To acquire the values, a calibration curve was constructed using known concentrations of ethanol vapor. A 1-liter airtight bottle was used as a test chamber, into which measured volumes of 100% undenatured ethanol ranging from 0.1 mL to 10 mL were injected and left for 2 minutes to allow for a uniform vapor distribution. These quantities correspond to BrAC values from 0.1 mg/L to 10 mg/L. The output voltage was recorded for each concentration, and the corresponding RSRO values were calculated based on the output voltage readings; the data collected was then used to determine the values for variables a and b (data, sample calculations and final BrAC formula will be displayed in the results section). Following the estimation of alcohol vapor concentration, the blood alcohol concentration (BAC) was calculated as a range by multiplying the BrAC value by 2.1 (BrAC X 2.1) and using the

Widmark formula. To estimate the grams of alcohol consumed, the user was asked to input the volume of the drink (in mL), the alcohol by volume (ABV) percentage, body weight (in kg), gender, and time elapsed since consumption (in hours). The mass of ethanol consumed was estimated using the formula [2]:

$$\text{Alcohol Mass (g)} = \text{volume (mL)} \times \left(\frac{\text{ABV}}{100} \right) \times 0.789$$

Where 0.789 g/mL is the density of ethanol. The BAC was then calculated using [10]:

$$\text{BAC (\%)} = \left(\frac{\text{Alcohol Mass (g)}}{\text{Body Weight (kg)} \times r} \right) \times 100 - (0.015 \times \text{Hours})$$

Where r is the alcohol distribution ratio, which was taken as 0.68 for males and 0.55 for females. The factor 0.015 represents the approximate rate at which the human body metabolizes alcohol per hour. With the two methods of calculating BAC, a secondary experiment was run with two participants consuming known quantities of alcohol in order to measure the difference between the two BAC values calculated, through which the accuracy of the breathalyzer may be illustrated.

In this LabVIEW program, a breathalyzer system is implemented to measure Blood Alcohol Content (BAC) based on real-time sensor readings and user input. The analog voltage from the MQ-3 alcohol sensor is read through an Arduino add-on using the LINX interface. The sensor voltage is converted into a resistance ratio (R_s/R_0), which is then used in a calibration formula to estimate alcohol vapor concentration (BrAC) in mg/L. Simultaneously, the user inputs values such as alcohol volume, alcohol percentage, body weight, sex constant, and time since drinking, which are used to compute BAC via the Widmark formula. Both the sensor-based BrAC and the Widmark-estimated BAC are displayed on the front panel. The program loops continuously until the stop button is pressed, updating values every second to provide real-time monitoring of alcohol exposure and estimated intoxication level.



Figure 2: Front panel of Labview

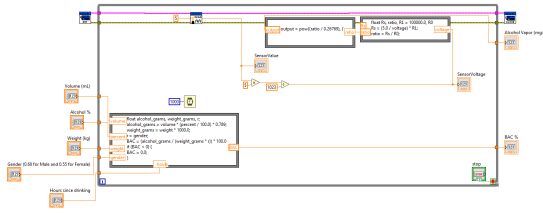


Figure 3: Labview block diagram

Code used in formula node boxes within Labview:

For Alcohol vapor in mg/L:

float Rs, ratio, RL = 100000.0, R0 = 200000.0;

$Rs = (5.0 / \text{voltage}) * RL$;

$\text{ratio} = Rs / R0$;

$\text{output} = \text{pow}((\text{ratio} / 0.26769), (1 / -1.5723))$;

Code used for BAC Calculation:

float alcohol_grams, weight_grams, r;

$\text{alcohol_grams} = \text{volume} * (\text{percent} / 100.0) * 0.789$;

$\text{weight_grams} = \text{weight} * 1000.0$;

$r = \text{gender}$;

$\text{BAC} = (\text{alcohol_grams} / (\text{weight_grams} * r)) * 100.0 - 0.015 * \text{hours}$;

if (BAC < 0) {

BAC = 0.0;

III. RESULTS

Using the calibration process described earlier, the resulting R_s/R_0 values were plotted against the BrAC results and have been recorded in Table 1, which can be seen in the appendix.

Log10(rs/ro) vs. Log10 BrAC

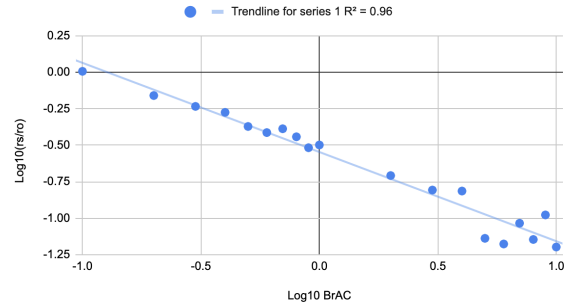


Figure 4: Log 10(Rs/R0) Vs Log 10 (BrAC) graph

The sensor output was recorded for each concentration, and the corresponding RSRO values were calculated based on voltage readings. The resulting data were used to plot a calibration curve with BrAC (mg/L) on the x-axis and RSRO on the y-axis. Upon log-transforming the data, a nearly linear relationship was observed, consistent with power-law behavior. With this, the slope value was found to be -1.572, which corresponds to the value of variable b, and the y-axis intercept was found to be 0.267, which corresponds to the value of variable a in the power-law formula $\log(RSRO) = \log(a) + b \log(BrAC)$, allowing us to determine that the relationship between BrAC and the resistance ratio:

$$BrAc = \left(\frac{R_s}{R_0} \right)^{\frac{1}{b}} = \left(\frac{\text{resistance ratio}}{0.267} \right)^{\frac{1}{-1.572}}$$

A log-log regression yielded a R2 value of 0.96, indicating that the ethanol concentration could explain 96% of the variation in sensor response. This strong correlation validates the accuracy and reliability of the breathalyzer circuit within the tested range. With this relationship, the circuit was further tested with 2 participants who consumed three alcoholic beverages that were 473ml each and 7% alcohol by volume; participant 1 was 95kg and participant 2 was 84kg, and their BrAC and BAC

(through the Widmark formula) values were measured after each hour. The BrAC value was then multiplied by 2.1 as discussed within the methodology to measure the difference between BAC calculated through the Widmark formula compared to BAC calculated through BrAC. Through table 2 (appendix) the low percent differences between the BAC values calculated through the Widmark formula and those estimated from measured BrAC indicate that the breathalyzer circuit is reasonably accurate. Across all recorded time points and participants, the percent differences remained within a narrow range of approximately 2% to 5%, suggesting that the sensor's output closely aligns with theoretical expectations. This level of agreement supports the validity of the calibration method used and demonstrates the circuit's capability to estimate BAC based on breath alcohol concentration reliably.

IV. DISCUSSION

This project explored the design and validation of a low-cost breathalyzer system using the MQ-3 alcohol sensor and an Arduino microcontroller. Beyond simply demonstrating functional output, the project aimed to evaluate the system's signal performance, medical relevance, environmental footprint, and economic feasibility. Through calibration, testing, and analysis, several practical considerations emerged across these four domains, each of which contributed to the strengths and limitations of the device as a potential self-monitoring tool for alcohol consumption. From a signal perspective, the system delivered consistent and interpretable output once calibrated, but not without challenges. The analog voltage signal from the MQ-3 sensor is inherently susceptible to electrical noise, especially in breadboard-based circuits where shared ground lines and jumper wires can introduce instability. Additionally, the MQ-3 sensor displayed signs of saturation after repeated use in short succession, producing elevated R_s/R_o ratios even in clean air. This required a waiting period between tests to allow the sensor to recover, which limited testing efficiency. Despite these signal-related challenges, the calibration curve achieved an R^2 value of 0.96, demonstrating a strong correlation between the

sensor's resistance ratio and known BrAC levels. This suggests that, when allowed to stabilize and used in a ventilated environment, the MQ-3 can provide a reliable analog input for estimating blood alcohol content.

From a medical standpoint, the implications of accessible breathalyzer technology are substantial. Breathalyzers serve as a critical first-line tool in both preventative medicine and impaired-driving enforcement, offering a quick, non-invasive method for estimating intoxication. By enabling individuals to estimate their BAC in real-time using inexpensive and customizable hardware, this project highlights the potential for breathalyzers to become a more common personal safety device. In contrast to commercial models, which can cost well over \$100 CAD and are often restricted to law enforcement or clinical use, this system can be built for a fraction of the cost and configured to account for user-specific factors such as weight, sex, and time since drinking. Throughout the experiment, BAC values estimated from the sensor-derived BrAC measurements were within 2% to 5% of those calculated using the Widmark formula. This accuracy suggests that such a device could help users make safer decisions around drinking and driving, and more broadly promote responsible alcohol consumption. While not intended for legal or diagnostic use, the breathalyzer's medical utility lies in its ability to empower personal awareness, particularly in young adults, students, or those in rural areas where access to medical-grade devices may be limited.

The environmental impact of the system was also considered. The breathalyzer circuit operates at a low power level (5V) and uses components like tin dioxide-based sensors and reusable electronics that minimize waste. Unlike single-use test strips or chemically reactive systems, the MQ-3 sensor and Arduino board are designed for repeated use. That said, the MQ-3 sensor includes an internal heating element that requires continuous current and can result in unnecessary energy consumption if left powered for extended periods. This issue was encountered during testing, particularly when the sensor was left on between participants, and could be addressed in future iterations through the implementation of a timed shutoff or sleep mode.

Additionally, while the alcohol vapors used during calibration were minimal, proper ventilation was essential to ensure safe indoor operation. Overall, the device offers a low-waste, energy-efficient alternative to commercial breath-testing systems when used responsibly.

Economically, the breathalyzer is highly accessible. The total cost of the prototype was approximately \$30 CAD, including a \$20 Arduino Uno, a \$5 MQ-3 sensor, and about \$5 in additional components such as resistors, wires, and a breadboard. All materials were readily available online or through local electronics stores, and the open-source nature of the Arduino platform makes replication straightforward for other users. This low barrier to entry is particularly valuable in educational settings, where the system could be used to teach students about analog signal processing, sensor calibration, and the physiology of alcohol metabolism. From a community health perspective, distributing affordable breathalyzers — even in kit form — could promote safer decision-making around alcohol use at a population level. Moreover, the project demonstrates that accurate physiological monitoring devices do not need to be expensive or proprietary to be effective. In summary, the breathalyzer developed in this project performed reliably across technical, medical, environmental, and economic metrics. While there are limitations — including sensor saturation and sensitivity to testing conditions — the system as a whole showed strong potential for practical application. With careful calibration, real-time input handling, and thoughtful design considerations, such a device could contribute to harm reduction, education, and personal safety in meaningful ways.

V. CONCLUSIONS

In conclusion, this project successfully developed and evaluated an Arduino-based breathalyzer prototype using an MQ-3 sensor for breath alcohol concentration (BAC) estimation, achieving reliable performance consistent with its design objectives. Through careful sensor calibration and testing, the device's BAC readings were found to be within approximately 2–5% of the values predicted by the Widmark formula; furthermore, the R^2 value of the

calibration curve plotted was 0.96, demonstrating a high level of accuracy relative to theoretical expectations. Additionally, the hands-on nature of this project provided the student team with invaluable experience in circuit design, Arduino microcontroller programming, and LabVIEW integration, effectively bridging theoretical coursework with practical biomedical engineering skills. Looking ahead, improving the device's energy efficiency and long-term reliability could further enhance its real-world utility, and its low-cost design positions it as a promising candidate for broader use as a public safety tool for alcohol monitoring.

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Appendix

Table 1: Experimental Data for Participants

Participant	Time (Hours)	Sensor Voltage (V)	Measured BrAC (mg/L)	BAC (%) through measured BrAC	BAC (%) through Widmark formula	Percent difference of measured BAC
1	1	2.08	0.0518	0.10879	0.10632	2.32%
	2	1.99	0.0454	0.09524	0.09132	4.29%
	3	1.88	0.0376	0.07902	0.07632	3.53%
2	1	2.18	0.0606	0.12734	0.1222	4.20%
	2	2.09	0.0525	0.1103	0.1072	2.88%
	3	1.99	0.0449	0.09421	0.0922	2.17%

Table 2: Calibration curve of BrAC compared to Rs/Ro values and Log Transformed

BrAC (mg/L)	Rs/Ro (simulated)	Log10 BrAC	Log10(rs/ro)
0.1	1.0142	-1	0.00612360621
0.2	0.6923	-0.6989700043	-0.1597056684
0.3	0.5832	-0.5228787453	-0.2341824847
0.4	0.531	-0.3979400087	-0.2749054789
0.5	0.425	-0.3010299957	-0.3716110699
0.6	0.3859	-0.2218487496	-0.4135252214
0.7	0.41	-0.15490196	-0.3872161433
0.8	0.3612	-0.09691001301	-0.4422522584
0.9	0.304	-0.04575749056	-0.5171264164
1	0.3174	0	-0.4983930776
2	0.1959	0.3010299957	-0.707965564
3	0.1559	0.4771212547	-0.8071538848
4	0.1535	0.6020599913	-0.8138916202
5	0.0728	0.6989700043	-1.137868621
6	0.0666	0.7781512504	-1.176525771
7	0.0924	0.84509804	-1.034328029
8	0.0715	0.903089987	-1.145693958
9	0.1053	0.9542425094	-0.9775716288
10	0.0635	1	-1.197226275

Arduino IDE code:

Microcontroller code (Arduino UNO through COM3 port)

```
const float R0 = 10000.0;
const float RL = 200000.0;
```

```
float userBAC = 0.0;
```

```
void setup() {
```

```
Serial.begin(9600);  
delay(500);
```

```
Serial.println("MQ-3 Breathalyzer - BAC Estimator");  
Serial.println("-----");
```

```
Serial.print("Enter volume of alcohol consumed (mL): ");  
while (!Serial.available()) {}  
float volume = Serial.parseFloat();  
Serial.readStringUntil('\n');  
Serial.println();
```

```
Serial.print("Enter alcohol percentage (%): ");  
while (!Serial.available()) {}  
float percent = Serial.parseFloat();  
Serial.readStringUntil('\n');  
Serial.println();
```

```
Serial.print("Enter body weight (kg): ");  
while (!Serial.available()) {}  
float weight = Serial.parseFloat();  
Serial.readStringUntil('\n');  
Serial.println();
```

```
Serial.print("Enter gender (M/F): ");  
while (!Serial.available()) {}  
String genderStr = Serial.readStringUntil('\n');  
char gender = 'M';  
if (genderStr.length() > 0) {  
    gender = toupper(genderStr.charAt(0));  
}  
Serial.println();
```

```
Serial.print("Enter hours since drinking began: ");  
while (!Serial.available()) {}  
float hours = Serial.parseFloat();  
Serial.readStringUntil('\n');  
Serial.println();  
float r = (gender == 'F' ? 0.55 : 0.68);  
float alcohol_grams = volume * (percent / 100.0) * 0.789;  
float weight_grams = weight * 1000.0;  
userBAC = (alcohol_grams / (weight_grams * r)) * 100.0;
```

```

userBAC -= 0.015 * hours;
if (userBAC < 0) {
    userBAC = 0.0; // BAC can't be negative
}
Serial.println("Calculating... Please wait for readings.");
Serial.println("Blow into the sensor when prompted to see changes.");
Serial.println();
}
void loop() {
    int sensorValue = analogRead(A5);
    float sensorVoltage = sensorValue * (5.0 / 1023.0);
    float Rs;
    if (sensorValue == 0) {
        Rs = 1e6; // avoid division by zero
    } else {
        Rs = RL * (1023.0 / sensorValue - 1.0);
    }
    float ratio = Rs / R0;
    float alcoholMgL = 0.189563503 * ratio * ratio
        - 8.617766543 * ratio
        + 1.079213151;
    if (alcoholMgL < 0) {
        alcoholMgL = 0;
    }
    float BAC = userBAC;
    Serial.print("Sensor voltage: ");
    Serial.print(sensorVoltage, 2);
    Serial.print(" V, Alcohol vapor: ");
    Serial.print(alcoholMgL, 3);
    Serial.print(" mg/L, Estimated BAC: ");
    Serial.print(BAC, 3);
    Serial.println("%");

    delay(10000); }

```